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Douglas Hofstadter and the Fluid Analogies Research Group, *Fluid Concepts and Creative Analogies: Computer Models of the Fundamental Mechanisms of Thought*, New York: Basic Books, 1995, ix + 518 pp., \$30.00 (cloth), ISBN 0-465-05154-5.

This is an important book, fizzing with ideas, promise, and achievement. Arising from two decades of innovative AI research by Douglas Hofstadter and his students, it challenges many assumptions of workers in – and critics of – both classical and connectionist AI. Hofstadter's computer models are intriguingly different from most others (though similar in spirit to some of the very earliest AI work, namely ANALOGY (Evans 1968) and ARGUS (Reitman 1965), as he remarks in Preface 6, "Two Early AI Approaches to Analogy"). In particular, they focus on an aspect of human thinking often claimed by post-Wittgensteinian philosophers to lie beyond the scope of any possible computational approach: its creative capacity to extend or reconfigure concepts endlessly, as the context changes.

The half-dozen programs described in these 500 pages can be thought of as models of analogy, or creativity. As such, some people might regard them as studies of an optional extra, a desirable feature of the mind to which psychologists can eventually turn after they have discovered how concepts are used in everyday thought. Hofstadter disagrees. For him, the creative extension and interrelation of concepts are fundamental aspects of thought, the rule rather than the exception. "Normal" and "creative" thinking differ in the extent to which concepts are allowed to "slip" from their normal constraints so that some relatively far-fetched similarity should be recognized. Hofstadter's work thus focusses not on creative analogy considered as a special case, but on conceptual thinking in general.

The central theme is that concepts are intrinsically fluid, or changeable, and that computer models claiming psychological significance should reflect this. Conceptual fluidity allows "conceptual slippage", wherein the mind applies pre-existing concepts to novel situations in ways that respect the particularities of the current context and that involve new ways of perceiving (describing) contextual items.

Another major theme – or, rather, another way of expressing the same point – is the ongoing reciprocal interaction between high-level perception and concepts. As Hofstadter puts it, “cognition is recognition” (p. 97). Our perception can be significantly altered by analogical thinking, as the initial (perceptual) description may be adapted, or even destroyed, during the recognition of the analogy. (Hofstadter compares this process with conceptual revolutions in science: The initial interpretation is discarded, and a fundamentally different interpretation is substituted for it.)

Most current computer models of analogy focus on the retrieval and mapping of pre-existing concepts, rather than the creative construction of new ones. No fundamental restructuring of concepts takes place. Hofstadter, by contrast, insists that analogy involves, not only comparisons between, but (often) also restructuring of previously existing concepts. He criticizes analogy models such as the Structure-Mapping Engine (SME; Falkenhaimer, Forbus, and Gentner 1990) for relying on (formal, meaningless) structural features that are carefully pre-assigned so that the intended conceptual mapping should take place. Moreover, their representations of concepts are fixed, remaining unaltered after the analogy has been drawn. Similarly, scientific-discovery programs such as BACON (Langley et al. 1987) use ready-made concepts and principles of inference provided by the programmer, so model conscious (and unadventurous) scientific reasoning rather than novel analogical insights. His own approach is significantly different.

Hofstadter’s group have implemented a set of parallel-processing systems including the early Jumbo (which built word-like letter-sequences out of jumbled groups of letters) and the more recent Copycat (which finds analogies between alphabetic letter-strings) and Tabletop (which compares actions directed at various layouts of eating-utensils). The wide range of analogies found by these programs have a “human feel” to them, and are sometimes both non-obvious and intuitively compelling. (For instance: If ABC goes to ABD, what does XYZ go to? Copycat, like people, offers various different answers, including the boring XYD and the more creative WYZ.)

Their novel architecture lies somewhere between classical AI on the one hand and connectionism on the other. Accordingly, they achieve types of computation prominent in the former, such as top-down processing and manipulation of hierarchical structure, as well as types more characteristic of the latter, such as bottom-up processing and multiple constraint satisfaction. In terms of my own distinction between “combinatorial” and “exploratory-transformational” creativity (Boden 1990),¹ Hofstadter’s programs at present model the first more closely than the second. But hierarchical structures and top-down guidance are allowed for, and stylistic constraints might be represented by “clamping” certain concepts or conceptual neighbourhoods more or less firmly. Hofstadter’s general architecture could therefore model the more systematic, sometimes deliberate, exploration and transformation involved in stylistic (as opposed to combinatorial) creativity.

These programs are nondeterministic, combining a multitude of competitive bottom-up processes with (changing) top-down influences. Hofstadter uses the term ‘terraced scan’, a parallel (and layered) breadth-first approach first explored many years ago in Jumbo (Chapter 2, “The Architecture of Jumbo”, by Hofstadter) and developed more fully in Copycat (Chapter 5, “The Copycat Project: A Model of Mental Fluidity and Analogy-making”, by Hofstadter and Melanie Mitchell) and Tabletop (Chapter 8, “Tabletop, BattleOp, Ob-Platte, Potelbat, Belpatto, Platobet”, by Hofstadter and Robert French).² The lowest-level processes concern tiny details. But, as processing proceeds, patterns of low-level activity develop that in effect act as larger-scale agents. The program simultaneously considers many different descriptions, on various levels, of its concepts or perceptual inputs. Its representations are built up dialectically, each step being influenced by (and also influencing) the type of analogical mapping that the current context seems to require. If a part-built interpretation seems to be mapping well onto the developing analogy, it is maintained and developed further; otherwise, it is dropped, and an alternative is tried, using different features of the target concept.

The range of possible interpretations depends on the descriptors available. Some mark relatively superficial features; others concern the inner structure of a concept or conceptual neighbourhood. The closeness of the match between the potential analogy and its target is assessed for each set of descriptors, and the probability of focussing on a particular match varies – and may be influenced by top-down guidance. These models can therefore draw analogies in a fairly rigid way or in a more creative fashion, by allowing less or more conceptual slippage, respectively. And they can recognize the difference: Copycat “knows” that, in the letter puzzle given above, the answer WYZ is more interesting – less predictable, more abstract, more highly-structured – than XYD.

Borrowing Piaget’s terminology, we could say that conceptual fluidity allows both assimilation of novelty and accommodation to it. However, these polysyllabic Piagetian terms are notoriously vague. Most discussions of analogy that show any significant subtlety and insight are vague likewise. Cognitive scientists who have tried to banish this vagueness by expressing their theories as AI models have usually sacrificed the subtlety, too. But this subtlety is not a desirable penumbra of thought, which with application – not to mention further funding – could be added to the core model later. Rather, it is the very heart of the matter. In short, most AI models of thinking (analogy, problem-solving, memory) throw out the baby of conceptual fluidity with the bath water of hand-waving vagueness.

To be sure, Hofstadter sometimes waves his hands, too. In particular, his exciting “Letter Spirit” project (Chapter 10, “Letter Spirit: Esthetic Perception and Creative Play in the Rich Microcosm of the Roman Alphabet”, by Hofstadter and Gary McGraw) is largely programmatic rather than programmed. The current implementation of Letter Spirit recognizes alphabetic letters in various fonts. This is the first step within an ambitious long-term project to implement the creative design of a very wide range of letter fonts. Each letter must resemble not only all

its cross-font cousins, all Bs for instance, but also 25 of its own-font siblings – constraints that are complex, subtle, and to some extent in conflict.

Just what counts as handwaving is, of course, a controversial matter. In particular, there may be – and in this area there is – disagreement over the degree to which a certain AI model or methodology is generalizable to other, more complex, cases. At various places in the book, Hofstadter points out that what AI workers describe as a ‘real-world problem’ may be no such thing, the appearance of “reality” being illusory. His argument is summarized in Preface 4, “The Ineradicable Eliza Effect and Its Dangers”: Most AI problem-solvers and analogy-programs merely match abstract syntactic structures, with no serious attempt to model the inner structure or external associations of the concepts apparently involved. If they seem to pass the Turing Test (discussed in the Epilogue, “On Computers, Creativity, Credit, Brain Mechanisms, and the Turing Test”), this is due merely to people’s over-generous interpretation of their natural-language front ends. So far as these programs are concerned, the concept slots might as well be filled by meaningless tokens like ‘X’ or ‘blurg’ as by English words such as ‘heat’ or ‘flow’. This is not the Searlean philosopher’s claim that no AI program, in principle, could ever understand anything. Rather, it is an AI practitioner’s claim that certain AI methodologies and architectures are closer than others to modelling the richness of human understanding.

He admits that his chosen domains are mere toy-worlds: Letter sequences and table layouts are not the stuff of drama, or of science. The laws of physics “discovered” by BACON, or the “real-world” analogies generated by SME, may seem more lifelike and impressive. But that, he argues, is the Eliza effect. Nothing truly concept-like is going on in these programs: All the intelligence, creativity, and conceptual thought lies outside, in the minds of the designers. By contrast, claims Hofstadter, Copycat and Tabletop work in psychologically realistic ways within their artificially restricted worlds. Their processing does not match human mental complexity, of course – but it captures something of the “essence” of conceptual thought.

This book merits the attention of everyone interested in cognitive science, critics and sympathizers alike. You, gentle reader, as a devotee of *Minds and Machines*, should certainly read it. If 500 pages seems too much of a good thing, then concentrate on the chapters describing Copycat and Letter Spirit, and the critique of classical AI developed in Chapter 4 (“High-level Perception, Representation, and Analogy: A Critique of Artificial-intelligence Methodology”, by David Chalmers, Robert French, and Hofstadter). If you have time only for one chapter, then concentrate on Copycat. But this time-efficient strategy would save your time at a cost. Hofstadter’s writing is crammed with acute phenomenological observations, intriguing comparisons, fascinating asides, and challenging speculations. It would be a pity to miss them.

Notes

¹Editor's note: See Shanker, Stuart (1993), Review of Boden 1990, *Minds and Machines* 3, pp. 109–113.

²Editor's note: See Shelley, Cameron, Review of Robert M. French (1995), *The Subtlety of Sameness: A Theory and Computer Model of Analogy-Making*, in *Minds and Machines* 7, pp. 292–296.

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Robert Cummins and John Pollock (eds.), *Philosophy and AI: Essays at the Interface*, Cambridge, MA: MIT Press, 1991, xi + 304 pp., \$29.95 (cloth), ISBN 0-262-03180-9; \$15.00 (paper), ISBN 0-262-53135-6.

Philosophy and computer science are both concerned at least in part with abstract symbolic systems, so it should come as no surprise that they have much to gain from one another. In particular, the interface between artificial intelligence (AI) and those branches of philosophy concerned with cognition ought to be especially fruitful. It is this interface that *Philosophy and AI* endeavors to explore.

AI has changed drastically in the six years since this book was published, and much earlier work has not aged well. While most of the technical work in the book is by now outdated, the philosophical implications of several essays are both deep enough to stand the test of time and sufficiently insightful to inspire new research. The book is certainly worth reading for these insights alone, despite the fact that much of the technical content is now far from the cutting edge.

The book is concerned almost exclusively with the relations between AI and analytic epistemology. This is a narrower focus than one ought automatically to expect. AI has sophisticated relations with ontology and descriptive metaphysics,